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Contents

Introduction	118
“Astronomy” and “Astrology”	118
Iatromathematics: Medical Astrology	119
Babylonian Astral Medicine	119
Astrological Melothesia	120
Decumbiture Charts	121
Astronomy in the Hippocratic Corpus: The Moon, Critical Days, and Humors	122
Crisis Periods	122
Seasons, Qualities, and Humors	123
The Influence of Galen and Pseudo-Galen	123
Humors, Temperament, and Astrology	125
Abū Ma’shar and Astrological Medicine	126
Judging Urine	126
Astronomy and Astrology in Medical Education	127
Physician-Astrologers of the Middle Ages, Renaissance, and Early Modern Period	128
Pietro d’Abano	128
Cecco d’Ascoli	129
Marsilio Ficino	129
Girolamo Cardano	129
Simon Forman	130
Nicholas Culpeper	130
Cross-References	130
References	131

Abstract

Astronomy and astrology were combined with medicine for thousands of years. Beginning in Mesopotamia in the second millennium BCE and continuing into

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the eighteenth century, medical practitioners used astronomy/astrology as an important part of diagnosis and prescription. Throughout this time frame, scientists cited the similarities between medicine and astrology, in addition to combining the two in practice. Hippocrates and Galen based medical theories on the relationship between heavenly bodies and human bodies. In an enduring cultural phenomenon, parts of the body as well as diseases were linked to zodiac signs and planets. In Renaissance universities, astronomy and astrology were studied by students of medicine. History records a long tradition of astrologer-physicians. This chapter covers the topic of astronomy, astrology, and medicine from the Old Babylonian period to the Enlightenment.

Introduction

In virtually all human cultures, physical connections have been noted between human bodies and celestial bodies. In western medicine, Hippocrates wrote of the influence of the moon on human bodies, and the system of humors he devised led to the development of temperament theory which was taken up in medicine and subsequently incorporated into astrology. In addition, astrology assigned parts of the body to planets and zodiac signs (*melothesia*). While astronomy records and predicts the positions of celestial bodies, it is through astrology – the interpretation of the effects of stars and planets on humans – that medicine incorporates astronomy into its practice. Thus, the principal focus of this chapter will be astrology’s connection with medicine; measurements provided by astronomy are relevant only insofar as they provide the means for astrological and medical interpretation.

“Astronomy” and “Astrology”

Until the time of Copernicus, the terms *astronomia* and *astrologia* were interchangeable (French 1994, pp. 33–34; Hübner 1989). In the second century CE, Claudius Ptolemy defined the two uses of *astronomia*:

Of the preparations for the goal of prognostication through astronomy, O Syrus, two are the greatest and most authoritative. One is first in order and in power, by which we comprehend for every occasion the configurations of the movements of the sun, moon, and stars which happen in relation to one another and to the earth. The second is by which, through the natural particular quality of the configurations themselves, we investigate the changes in the surrounding environment which they bring about. (*Tetrabiblos* I, 1, trans. Greenbaum [Ptolemy and Hübner 1998, pp. 3.31–39])

The first use is what we now call astronomy proper; the second is astrology. Ptolemy’s *Almagest* dealt with the first use; his *Tetrabiblos* (*Quadripartitum* in the Latin west) concerned the second, astrology. Ptolemy’s quotation encompasses both uses in the word “*astronomia*”. The subject of medicine necessarily deals with the diagnosis and treatment of living bodies, so when the words “astronomy”

or “astrology” are mentioned in a medical context in historical texts, both astronomy and astrology could be included in either term. In medicine, astrology became a resource for diagnosis and prescription.

Iatromathematics: Medical Astrology

No less an authority than Ptolemy highlighted the similarities between the practices of astrology and medicine (*Tetrabiblos* I, 3). Both use the very same word, *prognosis*, and both are considered “stochastic”, that is, conjectural, arts in which prediction is not always successful, but the goal is to do what is humanly possible for success (Greenbaum 2010).

Ptolemy further extolled the use of astrology in conjunction with medicine. He asserted that the Egyptians were experts in the combination of medicine with astronomy (*Tetrabiblos* I, 3). Furthermore, “iatromathematical systems” (*Tetrabiblos* I, 3) helped in understanding the kinds of commixtures and environments that affect bodies and disease, so that medicine can then deliver the correct treatment to each afflicted body. “Iatromathematics” is a compound word created from Greek *iatros*, doctor, and *mathêmatikos*, astrologer.

Iatromathematics in the west was practiced by the Babylonians, Egyptians, Greeks, and people of other Mediterranean cultures.

Babylonian Astral Medicine

Astrological medicine has a very long history in Mesopotamia (Heeßel 2008). As early as the Old Babylonian period (c. 2000–1600 BCE), healers would have medications “spend the night under the stars” (literally, “in the stars”) to gain astral efficacy (Reiner 1995, p. 48 and n. 188, pp. 49–55; Heeßel 2008, p. 3 and n. 10). Since stars could send disease – “The diseases *sikkatum*, *išātum*, *miqtum*, *šanudû*, *ašûm*, *sāmānum*, *epqennu*, *šalattinnu*, and *girgiššum* have come down from the stars of the sky” (Heeßel 2008, p. 2, citing Goetze 1955, p. 11) – it was only logical to take advantage of their influence for cures. Diseases were assigned to planets representing gods, as well as to constellations and the zodiac (Heeßel 2008, pp. 2, 8–9). Once the zodiac was introduced in the Persian period, early associations of illnesses with days of the month (hemerologies) were modified to apply to certain zodiac signs (Geller 2010b, p. 56). In Babylon, the *baru*-priest, associated with divination; the *asû*, physician; and the *āšipu*, ritual performer, were all involved with medical diagnosis and treatment. After ca. 500 BCE, the *baru*-priest used celestial configurations to make predictions.

Babylonian medico-zodiacal texts included the influence of the moon on both the type and course of a disease. The moon’s zodiac sign was also considered instrumental for performing efficacious rituals. The texts called “Stone, plant and tree”, dating from the fifth century BCE, were modified to include the zodiac and/or to apply to medical procedures (Heeßel 2008, pp. 9–12).

A very late Babylonian medical commentary assigned parts of the body to planets, in this case Jupiter to the spleen and Mars to the kidneys (Reiner 1995, pp. 59–60; Heeßel 2008, pp. 14–15; Geller 2010a, p. 158; Geller 2010b, pp. 60–80). This is the earliest example of what is called “melothesia”, assigning body parts to planets or the zodiac. This concept was taken up and further developed in Greco-Roman astrology.

Astrological Melothesia

While Babylon records the earliest example of melothesia, similar systems arose and developed within Greco-Roman astrology. In P. Michigan 149 (second century CE), portions of zodiacal signs are assigned to “front parts” and “back parts”, (Robbins 1936, p. 91)

Zodiacal melothesia was described by Manilius, Vettius Valens, Antiochus, and Rhetorius in the Greco-Roman period and Late Antiquity (Bezza 1995, II, pp. 722–731). They assigned the head to Aries, throat to Taurus, arms and hands to Gemini, breast to Cancer, heart and back (or sides) to Leo, belly to Virgo, kidneys to Libra, genitals to Scorpius, thighs to Sagittarius, knees to Capricorn, shins or ankles to Aquarius, and feet to Pisces. This arrangement persisted into the modern period. The famous 1416 miniature of zodiac man in *Les Très Riches Heures* of the Duc de Berry is a well-known example (http://commons.wikimedia.org/wiki/File:Anatomical_Man.jpg), but many others exist.

Astrological texts detailed the body parts, diseases, and injuries covered by each sign. Vettius Valens describes Aries in the following way:

Let Aries be said to signify generally those things concerning the head, the sense organs and eyes. And so the place makes . . . headaches, weak vision, apoplexy, deafness, dulling of vision, leprosy, skin eruptions, baldness, alopecia, becoming bald, lack of sensation, putrid sores, attacks on breathing, callouses, tumors and whatever is customarily associated with the sense organs, and with both the hearing and the teeth. (Vettius Valens 1986, *Anthology* II, 37.6–7, trans. Greenbaum)

Valens (II, 37.1–5) mentioned another unusual melothesia based on the Lots of Fortune and Daemon (predictive positions in a chart), which he claimed was the system of “the ancients”. Planetary melothesia is attested from this period and was also carried through to the modern period. In this system, planets and luminaries, beginning with the Sun and Moon, ruled over parts of the body. Again, Valens’ *Anthology* is a convenient source:

Sun: Of the parts of the body it rules over the head, sense organs, right eye, ribs, heart, respiratory or sensory movement, nerves [as organs of sensation]. . . (I, 1.2)

Moon: Of the parts of the body it rules over the left eye, stomach, breasts, bladder, spleen, membranes, marrow (from which it brings about those suffering from dropsy). . . (I, 1.5)

Saturn: Of the parts of the body, it rules over the legs, knees, sinews, watery fluids, phlegm, bladder, kidneys and parts hidden within. It is indicative of whatever injuries arise from cold and moisture, such as dropsies, tendinitis, gout, cough, dysentery, tumors, convulsions, daemonic possessions, unnatural lust, depravity. (I, 1.12–13)

Jupiter: Of the outer parts of the body it rules over the thighs, feet (from which it also supplies foot-racing in the athletic contests); of the inner parts, seed, womb, liver, parts on the right <and teeth>. (I, 1.18)

Mars: Of the parts of the body it rules over the head, rump, private part; of the inner parts, the blood, spermatic ducts, bile, excretion of excrement, hind-parts, moving backward, lying backward. (I, 1.24)

Venus: Of the parts of the body it rules over the neck, face, lips, organ of smell and the foreparts from foot to head, intercourse of the genitals, of the inner [parts], the lungs. (I, 1.33)

Mercury: Of the parts of the body it rules over hands, shoulders, fingers, joints, intestines, hearing, windpipe, bowels, tongue. (I, 1.42) (Vettius Valens 1986 all trans. Greenbaum)

These descriptions demonstrate the interrelationship between planetary and sign rulerships. For example Jupiter, the ruler of Sagittarius, rules the thighs, which are also assigned to Sagittarius. Venus (Taurus's ruler) and Taurus rule the neck. Mercury, which rules Gemini and Virgo, rules, with those signs, the hands, shoulders and fingers (Gemini), and intestines (Virgo). Thus, doctors using melothesia had a structurally coherent system for diagnosing and healing disease or injury. Astrological charts could be cast for the nativity of the patient, if the time and place were known, and/or for the time the patient took to his bed, a technique called "decumbiture".

Decumbiture Charts

Casting a chart for the time at which a patient became ill enough to go to bed also has a long tradition. The earliest securely attested literary use of this practice occurs in the *Apotelesmatika* of Hephaestio of Thebes in the early fifth century CE. However, the practice may well have been used earlier, for example, in healing centers such as that of Apollo Grannus at Grand, France, in the second century CE (Abry 1993), where astrological boards associated with client consultations have been found (Evans 2004). Using colored markers, the healer would have laid out on the board the planetary and ascendant positions for the time the illness began. Techniques for casting and interpreting a decumbiture chart appear in Book V, 41 of Dorotheus' *Pentateuch* (a.k.a. *Carmen Astrologicum*) (probably added at a later date). The tradition became a staple of astrological medicine, continuing into the medieval period and the Renaissance, and well into the seventeenth century (see, e.g., Nicholas Culpeper's *Astrological Judgement of Diseases from the Decumbiture of the Sick*, 1655).

The Moon was of primary importance in analyzing a decumbiture chart. As the *Pentateuch*, Book V, 41.1, 11 explains:

Whoever desires to know the condition of the patient [at] the beginning of when he is ill, how long he will endure [it], let him look at the ascendent and the Moon and the lord of the ascendent and the lord of the Moon's house and the Moon's conjoining with a star, whatever it is, and the Moon's dodecatemoron. . . . If you want to know the condition of the patient, when it will be heavy and when it will be light, then the indicator of this is the Moon. (trans. Pingree [Dorotheus and Pingree 1976])

In addition, the angles of the decumbiture chart represented the different players during the course of the disease:

Look concerning the matter of the patient from these four cardines [*i.e.*, *angles*]. The ascendent indicates the doctor, the midheaven indicates the patient, [the sign] opposite the ascendent indicates the cause of the illness, from what thing it is, and the cardine of the fathers [the fourth] indicates his recovery and his medication, if God wishes. (*Pentateuch* V, 41.35–36, trans. Pingree [Dorotheus and Pingree 1976])

Finally, planetary positions in the decumbiture could be compared with the natal chart: “If you want to know the condition of the patient and his death and he is one of those whose nativity is known, then look at the Moon, if it transits in its motion in the sign in which Saturn was in his nativity”. (*Pentateuch* V, 41.42, trans. Pingree [Dorotheus and Pingree 1976])

Astronomy in the Hippocratic Corpus: The Moon, Critical Days, and Humors

Possibly the first Greek evidence of heavenly bodies affecting health appears in *Regimen IV* (Dreams) of the Hippocratic corpus, where dreaming of the heavens gives indications for the dreamer’s health or illness (*Regimen IV*, 89, in Jones (1931), pp. 426–437).

The Moon was also an important component of Greek medical diagnosis. In the corpus of treatises associated with Hippocrates, its phases (which created the month) coincided with “critical days” of illness, the days in which medical crises occurred. Moon phases and critical days were integral parts of medical diagnosis and prescription. Also integral was the doctrine of humors, the identification of four bodily fluids associated with health and disease; these were linked to the seasons, and therefore had a solar connection. Humoral theory led to the development of temperament theory, the assignment of a bodily constitution based on the relationship between the four humors and their proportions in each person. Temperament (also called “complexion”) became important in medical treatment. All of these concepts – critical days, crises, and humors – were eventually, if not initially, incorporated into iatromathematical practice.

Crisis Periods

Acute diseases were judged by Moon cycles, while chronic diseases involved Sun cycles (French 2003, p. 132). The course and prognosis of an acute disease, and its crisis periods, often, though not always, corresponded with the phases of the Moon. By knowing the critical days and the circumstances surrounding them, doctors could better diagnose and treat illness. The underlying structure of critical days appears to be based on the week and month, and thus has a lunar component, although some schemes of critical days can be difficult to pinpoint to a part of the lunar cycle. Hippocratic Aphorism II, 23 states, for example: “Acute diseases come

to a crisis in fourteen days” (Jones 1931, p. 113), suggesting the Full Moon (opposition of sun and moon); but other days do not as clearly correspond to a lunar phase (e.g., Aphorisms II, 24). In his commentaries on Hippocrates, the second-century physician Galen linked critical days much more precisely and specifically to the phases of the Moon (Pennuto 2008, pp. 77–79).

Seasons, Qualities, and Humors

The Hippocratic *Humors* (Chap. 8) briefly relates humors and general body constitution to disease. Other Hippocratic works flesh out the theory of humors and its relationship to the seasons and qualities of hot, cold, wet, and dry. In *Aphorisms*, the seasonal change produced by the solar cycle brings about disease (Aphorism III, 1): “It is chiefly the changes of the seasons which produce diseases, and in the season the great changes from cold or heat, and so on according to the same rule”. (Jones 1931, p. 123) This emphasis on seasons and the observation that extremes of hot and cold produce disease combine to show how the qualities affect both disease and human constitutions. In the Hippocratic treatise *On the Nature of Man*, vii, the four basic humors, or fluids of the body, are in turn associated with a quality: yellow bile (*cholê xanthê*) with hot, black bile (*cholê melainê*) with dry, blood (*haima*) with wet, and phlegm (*phlegma*) with cold. *On the Nature of Man*, iv says:

The body of a man has in itself blood, phlegm, yellow bile and black bile; these make up the nature of his body, and through these he feels pain or enjoys health. Now he enjoys the most perfect health when these [things] are duly proportioned to one another in respect of compounding, power and bulk, and when they are perfectly mingled. (Jones 1931, p. 11)

The idea of a balanced proportion of humors in the body led to the development of temperament. A person’s basic temperament affects her response to changes in season, and to diseases associated with a particular quality or humor. Aristotle further developed the system of qualities and elements (*On Coming-to-be and Passing-away*, II). By the time Ptolemy wrote the *Tetrabiblos*, the concept of temperament had developed to the point that Ptolemy incorporated its determination into the structure of astrological practice, and it continued to be included in astrological medicine thenceforth. By knowing a temperament which could be discerned, at least partially, from the birthchart, a doctor could more accurately prescribe a particular treatment for a patient. The development of humoral theory and temperament in application to medical practice was expanded and popularized by Galen.

The Influence of Galen and Pseudo-Galen

The great Greek physician Claudius Galen (b. 131 CE) is arguably the greatest influence on western medicine before 1700. His massive corpus (over 20 volumes in Greek, plus many more translated into Arabic and Latin) was subsequently reproduced, translated, and supplemented. Many pseudo-Galenic works were

considered to be authentic in the Middle Ages and Renaissance, including a treatise on astrological medicine, *On Prognosis from Decumbiture*. Among authentic works are *On Temperaments*, *On Crises*, and *On Critical Days*. In the Middle Ages and Renaissance, an important text of medical astrology was *Aggregationes de crisi et de creticis diebus*, a synthesis of Galen's *On Crises* and *On Critical Days* (O'Boyle 1991).

Galen revered both Hippocrates and Aristotle, and his interpretations of their work colors the way in which they were perceived after Antiquity. The humoral model espoused by Galen became the paradigm in much medieval and Renaissance medicine. Astrological medicine, too, used the humoral model based on Aristotelian qualities and elements.

There is evidence that Galen endorsed the inclusion of some astrology within medicine. Though he disdained the more exuberant claims of astrologers to predict the events of an entire life from the birthchart (see Galen, *Commentary on Hippocrates' Airs Waters and Places*, III, 11) (Toomer 1985, p. 199, section 14; Cooper 1999, pp. 43–44), he acknowledged that astrological methods could be combined with medicine to treat disease. In this, he praised the Egyptians for their work on lunar effects, which he found through experience to be true (Barton 1994, p. 54 and n. 132; Nutton 2008, pp. 21–22).

Galen, like Ptolemy, looked for physical causes to explain the influence of the heavens on the human body. He gave the Hippocratic critical days theory a firm foundation in the phases of the Moon. Astrologically, *On Critical Days* stresses the Moon's importance in its relation to the zodiac, in conception and birth, and in the beginning of any medical event (Barton 1994, p. 54). Conception and birth were already important astrological concerns (Frommhold 2004) and the Moon was a major factor for interpretation in electional and horary charts (cast, respectively, for the times of events and questions).

In *On Temperaments*, Galen ordered and elucidated the meanings of the qualities in application to the humors. In other works such as *On the Elements According to Hippocrates*, *On the Natural Faculties*, and *On Hippocrates' On the Nature of Man* Galen developed a synthesis of humors, elements, and qualities:

“...that all things are generated from heat, cold, dryness and wetness, and for this reason these elements are common to all. Blood, phlegm, yellow bile and black bile are the particular elements of the nature of man. ... And it is quite clear that each of them comes from the four primary elements which we call wetness, dryness, cold and heat. ... So the hot, cold, dry and wet parts seen clearly in the human body, are not the elements of the nature of man, but compositions and generations of these elements; water, fire, air and earth”. (Galen et al. (trans.) *On Hippocrates' On the Nature of Man*, §§51–53, 54, trans. Lewis and Beach; see Greenbaum 2005, p. 16)

On the astrological front, Ptolemy also advanced the idea of temperament as something which could be determined from the birthchart by applying the qualities of hot, cold, wet, and dry to celestial bodies, the seasons and the chart itself. His arrangement was instrumental in promoting temperament within astrology.

Humors, Temperament, and Astrology

Claudius Ptolemy was a near contemporary of Galen (about 40 years his elder). Although he never used the word “humor” in the *Tetrabiblos*, he followed the doctrines of Hippocratic medicine in espousing a balance of qualities for good health. He considered temperament, a “commixture” (*synkhrasis*) of the qualities of hot, cold, wet, and dry in every person, to be discernible from the birthchart: “. . . why can he not also perceive, for each individual person, the general quality of his particular commixture from the surrounding environment at his creation (such as, that his body is such-and-such, and his soul such-and-such). . . ?” (*Tetrabiblos* I, 2.11, trans. Greenbaum)

Ptolemy assigned planets one or more qualities based on their intrinsic nature, or their orientality or occidentality. The seasons, solstices and equinoxes, angles of the chart, and moon phases were also given qualities, as in the table above (Table 8.1).

Tetrabiblos III contained Ptolemy’s instructions for delineating temperament from the birthchart, using the Ascendant (the sign rising on the eastern horizon at the time of birth), planets in the Ascendant sign and their rulers, and the Moon for the shape of the body. Fixed stars rising with the Ascendant were also incorporated to a lesser degree (*Tetrabiblos* III, 12; see Greenbaum 2005, pp. 18–20, 141). Other astrologers of Late Antiquity also assigned qualities and elements to the planets and zodiac signs, but except for Hephaestio (who quoted Ptolemy extensively), the use of astrological temperament was not discussed.

Nevertheless, the importance given to temperament in medicine by such renowned physicians as Galen, coupled with the use of astrology in medical practice, led to further development of humors, elements and qualities in astrology, primarily under Arabic-language astrologers of the early Medieval period. By the end of the Middle Ages, astrologers began to incorporate astrological formulae for temperament into their work.

Table 8.1 Ptolemy’s assignment of qualities (*Tetrabiblos* I, 2, 4, 5, 8, 10, 11)

	Angles	Solstice/ equinox	Moon phase	Planet phase	Planets	Seasons
Hot	Mid-heaven	Summer solstice	Full moon	Opposition	Sun, Venus, Mars, Jupiter, Moon (slightly)	Spring, summer
Cold	Imum Coeli	Winter solstice	New moon	Conjunction	Saturn	Autumn, winter
Wet	Descendant	Vernal equinox	First quarter	1st station/ square	Moon, Venus, Jupiter	Winter, spring
Dry	Ascendant	Autumnal equinox	Last quarter	2nd station/ square	Sun, Mars, Saturn	Summer, autumn

Abū Ma'shar and Astrological Medicine

Abū Ma'shar (787–886 CE) was a committed Aristotelian and successor of Ptolemy, in that he was concerned with finding a rational, physical basis for astrology. Like Ptolemy, he noted similarities between astrology and medicine, but considered astrology to be even greater than medicine because it was concerned not only with nature and the sublunar world, but with the entire universe, including the heavenly bodies and stars. His *Great Introduction to Astrology* gave his hierarchy of predictive professions, where medicine held the middle rank but astrology the highest, as it is “a lofty profession and its subject is the stars which do not alter and are not subject to coming-to-be and passing-away. . .” (I, 2.25; Burnett and Yano forthcoming, p. 29). Skilled doctors, he said, knew the value of astrology:

For these know the excellence of the science of astrology, and they know that the science of the stars is the principle of the science of medicine, and they use the art of astrology together with the art of medicine all the time, in order to know when pains increase and decrease, and when they should be treated. (I, 5.36; Burnett and Yano forthcoming, p. 97).

Astrology can tell whether a person is about to die and thus not benefit from treatment (determining life expectancy was a common astrological practice), and when crisis days in an illness will occur (here Abū Ma'shar cites Hippocrates and Galen) (I, 5.39; Burnett and Yano forthcoming, pp. 99, 101).

Abū Ma'shar codified the linking of humors to planets and signs (Tables 8.2 and 8.3):

The same system appears in §347 of al-Bīrūnī's (973–1048) *Book of Instruction in the Elements of the Art of Astrology* (al-Bīrūnī and Ramsay Wright 1934, p. 211). He stated that “the signs are also indicative of the various diseases of man, of his complexion, figure, face and the like. . .” (al-Bīrūnī and Ramsay Wright 1934, §359, trans. Ramsay Wright). Antonius de Montulmo's 1396 *On the Judgment of Nativities* (► Chap. 9, “Ancient “Observatories”: A Relevant Concept?”) provided a formula (based on Ptolemy's) for determining temperament and quality of soul from the birthchart. Such formulae were helpful not only for astrologers but also for physicians. Indeed, the development of such astrological formulae for temperament likely occurred in part because of their usefulness in medicine.

Judging Urine

A text from the Medieval period presents an odd intersection of astrology and medicine. Uroscopy was the examination of a patient's urine to give clues for diagnosis and prognosis (French 2003, p. 132). But in 1219, William of England wrote *De urina non visa*, which explained how to judge urine astrologically without even seeing it. This little book was taught by statute in the university at Bologna, (O'Boyle 1991, p. 5; French 1994, p. 45 and n. 43; Grendler 2002, p. 410) so it was not as evanescent as one might think.

Table 8.2 The nature of signs according to Abū Ma‘shar (Abū Ma‘shar et al. 1994, *Abbreviation of the Introduction to Astrology*, ► Chap. 1, “Concepts of Space, Time, and the Cosmos”)

Sign	Quality			Element	Humor
Aries	Hot	Dry		Fire	Choleric (yellow bile)
Taurus		Dry	Cold	Earth	Melancholic (black bile)
Gemini	Hot		Wet	Air	Sanguine (blood)
Cancer			Cold	Water	Phlegmatic (phlegm)
Leo	Hot	Dry		Fire	Choleric (yellow bile)
Virgo		Dry	Cold	Earth	Melancholic (black bile)
Libra	Hot		Wet	Air	Sanguine (blood)
Scorpius			Cold	Water	Phlegmatic (phlegm)
Sagittarius	Hot	Dry		Fire	Choleric (yellow bile)
Capricorn		Dry	Cold	Earth	Melancholic (black bile)
Aquarius	Hot		Wet	Air	Sanguine (blood)
Pisces			Cold	Water	Phlegmatic (phlegm)

Table 8.3 The nature of planets according to Abū Ma‘shar (Abū Ma‘shar et al. 1994, *Abbreviation of the Introduction to Astrology*, ► Chap. 5, “Astronomy and Power”)

Planet	Quality			Element	Humor
Sun	Hot	Dry			
Moon			Cold	Wet	
Venus			Cold	Wet	Phlegmatic (phlegm)
Mars	Hot	Dry		Fire	Choleric (yellow bile)
Jupiter	Hot		Wet	Air	
Saturn		Dry	Cold		Melancholic (black bile)

Astronomy and Astrology in Medical Education

The universities of Paris, Montpellier, Bologna, and Padua were well-known centers for medical education beginning in the late twelfth and early thirteenth centuries. Within the university curricula, medicine was closely tied to the study of the arts; medical students were required, especially at universities with prominent medical faculties, to be conversant in logic, natural philosophy, astrology, basic mathematics, and astronomy (Siraisi 1990, pp. 66–67). Because of the traditional link between medicine and astrology, courses in medical education emphasized the teaching of astrological techniques applied to medical practice. In fact, “some measure of astrological competence was indeed one of the marks separating an educated practitioner from an empiric” (Siraisi 1990, p. 68).

At Bologna, for example, future doctors were required to study philosophy and astronomy-astrology for three years each; and medicine and its associated practice each for four years (Siraisi 1990, p. 72). The astrological curriculum at Bologna, mandated by statute in 1405, shows the texts studied over a four-year period (Grendler 2002, pp. 410–412):

1st year: an *Algorismus* on fractions and integers (possibly by Sacrobosco)

- Euclid, *Elements*, Book I, with Campanus of Novara's *Commentary*
- *Alfonsine Tables* with *Canons* (probably John of Saxony's version)
- Anonymous, *Theory of the Planets*

2nd year: Sacrobosco, *De sphaera*

- Euclid, *Elements*, Book II
- Jean de Linières, *Canons* on the *Alfonsine Tables*
- Māshā'allāh (Messahalla), *Treatise on the Astrolabe*

3rd year: Al-Qabīṣī (Alcabitius), *Introduction to the Art of Astrology*

- (pseudo-)Ptolemy, *Centiloquium* with 'Alī ibn Riḍwān's (Haly Rodohan) commentary
- Euclid, *Elements*, Book III
- "Treatise on the Quadrant"

4th year: Ptolemy, *Quadripartitum* (*Tetrabiblos*)

- *De urina non visa*
- Ptolemy, *Almagest*, "third part" (possibly Book 3 on solar theory)

The medical curriculum at Bologna included Avicenna's *Canon*; commentaries on Hippocrates; and Galen's works on critical days, crises, fevers, simple medicines, and temperaments.

Physician-Astrologers of the Middle Ages, Renaissance, and Early Modern Period

Given the importance of astronomy and astrology in medical training, it is not surprising that a number of physicians were also accomplished astrologers – or conversely, that astrologers could also be accomplished physicians. Their combined skills in medicine and astrology enhanced the practice of both disciplines (and university-trained doctors were in demand to serve as court physicians). Some examples will serve to illustrate this genre of practitioners.

Pietro d'Abano

In the thirteenth and early fourteenth centuries, Pietro d'Abano (d. ca. 1315) was a vigorous proponent of astrology in general and of the use of astrology in medicine. He taught astronomy, astrology, philosophy, and medicine at Padua (French 1994, p. 55; Siraisi 1990, p. 135). He argued, as did Abū Ma'shar, that astrology is an important part of medicine: "No one should place himself in the hands of a doctor who is ignorant of astrology" (*Conciliator*, p. 4^r, cited in French 1994, p. 58). His scientific and astronomical knowledge was such that he could criticize Galen's theory of critical days on the grounds that Galen's knowledge of astronomy and astrology was insufficient to explain it properly (Siraisi 1990, p. 136). Differentia 10 of the *Conciliator* (1303) was wholly devoted to astrological medicine, including temperament and crises determined astrologically, melothesia, surgery based on astrological rules and astrological talismans ("*immagines*") (Vescovini 1987, pp. 30–32).

Cecco d'Ascoli

Cecco d'Ascoli (d. 1327), physician, poet, and astrologer, lectured on astronomy and astrology at Bologna in the early 1320 s; his *Commentary* on Sacrobosco's *Sphere* (before 1324) contained a summary of his lectures at Bologna, which had a strong astrological component, including justifications of astrology in medical practice (Siraisi 1990, pp. 67–68).

Marsilio Ficino

In the fifteenth century, Marsilio Ficino (1433–1499), the great Renaissance translator of Plato, studied medicine, theology, philosophy, and astrology in Florence. His *Three Books on Life* (Ficino et al. 1998) applied astrological methods to the treatment of melancholy; advice on diet and lifestyle in the first book; medicine in the second; and talismans, astrological magic, and music in the third all provided therapy for the melancholic. Such a lengthy treatise on melancholy demonstrates how important temperament theory had become to both life in general and medical practice. Ficino considered the reason for his own melancholic temperament to be the prominence of Saturn in his birthchart (Ficino 1997, p. 160).

Girolamo Cardano

Although there had always been opponents of astrology, in the sixteenth century questions about astrology's validity, and medical astrology in particular (most famously by the Italian humanist Pico della Mirandola), found more favor, leading to an erosion of belief in its efficacy in medical practice. This was the harbinger of astrology's eventual demise as a science in the Enlightenment and the privileging only of its sister astronomy. However, the physician and astrologer Girolamo Cardano (1501–1576) still valued astrology as a predictive science. Again like Abū Ma'shar, Cardano (in his *Commentary on Ptolemy's Tetrabiblos*) listed the arts which dealt with prognosis, and proclaimed “the noblest of these is astrology”, a statement he later qualified, however (Grafton and Siraisi 2006, p. 101, citing Cardano 1554, p. 1). Within medicine, though, he considered astrologically electing times for surgery to be a valid practice (Grafton and Siraisi 2006, p. 75, citing *Commentary on the Tetrabiblos*, 714). On occasion, he used event charts to give after-the-fact astrological analyses beginning with the time of illness or injury (Grafton and Siraisi 2006, pp. 69–71). He compared the practices of Hippocrates and Galen in determining natal temperament to astrologers determining the same from the birthchart (Grafton and Siraisi 2006, p. 104).

Simon Forman

Cardano's antithesis in the sixteenth century was the "notorious astrological physician of London", Simon Forman (Traister 2001). Forman (1552–1611) left historians a treasure trove: seven years of medical casebooks, in which meticulous notes and charts detailed his work with patients. Forman's formal medical training was nonexistent; though briefly a "poor student" at Oxford, he was unaccredited as a physician until 1603, when Cambridge gave him a license to practice physic and astronomy (Kassell 2006, p. 345). On numerous occasions, he fell afoul of the law. Astrology was crucial to his medical practice: He cast charts for the time of the patient's question about his illness, and then interpreted it for the disease and its outcome. Medically, he was a Galenist with Paracelsian leanings (Traister 2001, p. 51). However, his treatments were often brutal and repulsive, for example, the drinking and eating of excreta (Traister 2001, pp. 50–51). Forman wrote on the four humors, including a long section on melancholy, and may have treated Robert Burton (author of *The Anatomy of Melancholy*) (Traister 2001, p. 48). He practiced alchemy and astral magic, and dabbled in necromancy. He authored "The Astrological Judgements of Phisick and other Questions" in 1606, in which he explained how to cast and interpret horary charts within medical practice.

Nicholas Culpeper

The seventeenth century produced Nicholas Culpeper (1616–1654), a doctor-astrologer who trained as an apothecary and was a renowned herbalist. His sympathies for the common folk, whose circumstances prevented them for obtaining medical advice, led him to take up the practice of "physick" and write medical books in English, rather than the usual Latin. These included the never-out-of-print *Culpeper's Complete Herbal* (originally entitled *The English Physitian* [1652]); a treatise on temperament based on Galen's *Ars Medica* that he called *Galen's Art of Physick* (1652); and the above-mentioned *Astrological Judgement of Diseases from the Decumbiture of the Sick* (1655) (a title possibly derived from the pseudo-Galenic *On Prognostication from Decumbiture*).

By the eighteenth century, links between astrology and medicine grew weaker as astrology continued to decline in stature. Though tenuous connections between medical treatment and the phases of the moon continued into the 1700s (Fraser 1921), astrology was effectively written out of medicine by the nineteenth century.

Cross-References

- [Greco-Roman Astrology](#)
- [Late Babylonian Astrology](#)
- [Mesopotamian Celestial Divination](#)

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